

William Hager

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Education

B.S. in Computer Science, University of Rochester

Spring 2028

- GPA: X.XX/4.00
- Dean's Scholarship, All Semesters
- Bond Scholar, 2026 (Awarded to students who have an upcoming internship in the Upstate NY area)
- Coursework: Advanced Algorithmic Design, Introduction to AI/ML, Formal Computational Models and Limitations, Data Structures and Algorithms, Linear Algebra, Differential Equations

Work Experience

Undergraduate Research Assistant, Roc-HCI Lab

Fall 2025 - Present

- Contributed to development of [HAL \(Human Aligned LLM\)](#), a framework for **LLM alignment** with emphasis on inducing human-likeness traits
- Selected and curated training data for **DPO algorithm** to improve model realism, increasing win-rate on a human-likeness Turing test from 53% to 62%
- Developed and implemented [Turing-Bench](#), a **HuggingFace LLM Benchmark** that evaluates model submissions on ability to recognize if a generated conversation is human/AI or AI/AI
- Contributed to the *LLM Persuasion Judge*, an ongoing project working to develop persuasive skills in LLMs for the purpose of combating misinformation

Head Lifeguard, University of Rochester Goergen Athletic Center

Fall 2025 - Present

- Ensured pool patron safety through **clear communication** of pool rules as well as in-water rescues
- Managed a team of 35 other lifeguards, ensuring all scheduling and training demands were met
- Tracked and adjusted pool chemical levels

Projects

[Runesworn](#)

Spring 2025 - Present

- Designed and developed a **dice-rolling RPG video game** with Godot engine and GDScript (**Python**), using Godot's built in state machine and signal system to implement turn based combat and character building
- Created **database and data management system** to efficiently store and retrieve game and player data, including 12 unique encounters and 3 playable characters, as well as 12 unique dice
- Established and balanced core game mechanics with a small-scale playtesting group in order to ensure balanced and enjoyable gameplay, as well as dynamic difficulty scaling

[University of Rochester BAJA SAE Website](#)

Spring 2026

- Designed and built a custom full stack web application using a **React + Netlify Serverless + MongoDB** stack, averaging 20 weekly visits
- Implemented an Auth0 protected **schema-based CMS layer** to allow non-coding administrators to update dynamic content without code changes
- Built scalable backend services using **RESTful methods** to ensure longevity and stability

Mancala Engine

Spring 2026

- Implemented the **Minimax algorithm** to create an interactive mancala playing engine with a tuned evaluation heuristic and **100% win-rate** against average players
- Utilized optimization techniques like Alpha/Beta pruning to **decrease decision time by 99.1%** when examining trees with high depth
- Made use of iterative deepening techniques to limit response time to 500ms while still maintaining maximum nodes explored (>12 layers)

Skills

Programming Languages: Java, C, Python, GDScript, HTML, CSS, JavaScript

Frameworks: Java Swing, Java AWT, JavaFX, LibGDX, scikit-learn, React, Gradio

Technical Skills: Object-Oriented Programming, Linux/Unix, Valgrind, Full Stack Web Development, LLM Alignment